

ENGR 102 Summer Bridge

Day 15 Instructor Key

Append, Concatenation, Slicing, and Strings as Sequences

20 points • approximately 20-25 minutes

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Boolean-operator convention. Python operators appear as [and](#), [or](#), and [not](#). Their lowercase spelling is required in code.

Show intermediate state for trace credit. Program credit requires one coherent solution. Unless a prompt says otherwise, give exact Python 3 output.

Question 1. Trace sequence state and direct slicing.

4 points

```
text = "bridge"
pieces = []
for index in range(0, len(text), 2):
    pieces.append(text[index])
middle = text[1:5]
result = pieces + [middle]
print(pieces)
print(middle)
print(result)
```

Give the index ledger and every exact output line.

Answer and explanation

Indexes 0, 2, and 4 store b, i, and g, so pieces is ['b', 'i', 'g']. text[1:5] includes indexes 1 through 4 and is ridg. result is ['b', 'i', 'g', 'ridg']. These are the three printed lines.

Question 2. Repair sequence concatenation.

3 points

```
items = ["x"]
items = items + "y"
print(items)
```

Explain the failure and give both a concatenation repair and an append repair.

Answer and explanation

A list cannot concatenate directly with a string, so + raises TypeError. Concatenation repair: items = items + ['y']. Append repair: items.append('y'). Either produces ['x', 'y'].

Question 3. Program construction**13 points**

Write one complete Python program that satisfies every requirement.

- Read word as text and print invalid if its length is less than 3.
- Otherwise append characters at odd indexes to odd_characters.
- Count lowercase t characters.
- Build frame from the first one-character slice plus the last one-character slice.
- Print odd_characters, t_count, and frame. Include an iteration test with expected output.

Answer and explanation

```
word = input()

if len(word) < 3:
    print("invalid")
else:
    odd_characters = []
    t_count = 0
    for index in range(len(word)):
        character = word[index]
        if index % 2 == 1:
            odd_characters.append(character)
        if character == "t":
            t_count = t_count + 1

    frame = word[:1] + word[len(word) - 1:]
    print(odd_characters)
    print(t_count)
    print(frame)
```

For iteration, odd indexes 1, 3, 5, and 7 store t, r, t, and o, so odd_characters is ['t', 'r', 't', 'o']. There are two lowercase t characters. The first and last characters concatenate to in.

The validation branch prevents a short word from entering the sequence algorithm. The loop maintains list state and count state independently.

Scoring guidance

2 validation, 3 traversal/state, 3 odd-index collection, 2 t count, 2 slices/concateration/output, 1 complete iteration test.